



PARTS LIST

Semiconductors:

IC1	- ATmega328P-PU microcontroller
M1	- LM2596 buck converter module
LED1	- 3mm LED
D1	- 1N4007 diode
GSM	- SIM800L GSM module
GPS	- Neo-6M GPS module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 100-ohm
R2, R4	- 10-kilo-ohm
R3	- 1-kilo-ohm
R5	- 2-kilo-ohm

Capacitors:

C1	- 100 μ F, 35V electrolytic
C2, C3	- 22pF ceramic
C4	- 0.1 μ F ceramic

Miscellaneous:

X1	- 16MHz crystal oscillator
SW1, SW2	- Pushbutton switch
B1	- 12V battery or 12V DC adaptor
PZ1	- Piezo buzzer

respectively. The 3D view is shown in Fig. 7.

Assemble the circuit on a breadboard or PCB. Insert the SIM card in



Fig. 8: Author's prototype on PCB

the slot of GSM module. When SW1 is pressed, GSM calls the mobile number already entered in the program code and the buzzer sounds. The buzzer sound is to alert nearby people.

Once the call is established, GPS

gets latitude, longitude, and altitude coordinates and thus the exact location of the caller. The link of this location is sent to the receiver's mobile phone along with the recorded SMS message. On opening the received link, the receiver

gets to know the exact location of the caller on Google map. **EFY**



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